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**POLITECNICO**  
MILANO 1863

# Internet Of Things: Challenge 2

## Part 2

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# 1 Exercise

## 1.1 EQ1

Compute the total communication energy consumed by the battery-powered sensor and actuator over 1 hour in:

(a) CoAP (direct communication)

(b) MQTT (via the gateway)

Provide numerical results in  $\mu\text{J}$  (microjoule).

Energy Costs

$$E_{\text{tx}} = 50 \text{ nJ/bit}$$

$$E_{\text{rx}} = 58 \text{ nJ/bit}$$

CoAP message sizes

$$\text{CON PUT Request} = 70 \text{ B} = 560 \text{ bit}$$

$$\text{PUT ACK} = 15 \text{ B} = 120 \text{ bit}$$

MQTT message sizes

$$\text{Subscribe} = 65 \text{ B} = 520 \text{ bit}$$

$$\text{Sub Ack} = 55 \text{ B} = 440 \text{ bit}$$

$$\text{Publish} = 75 \text{ B} = 600 \text{ bit}$$

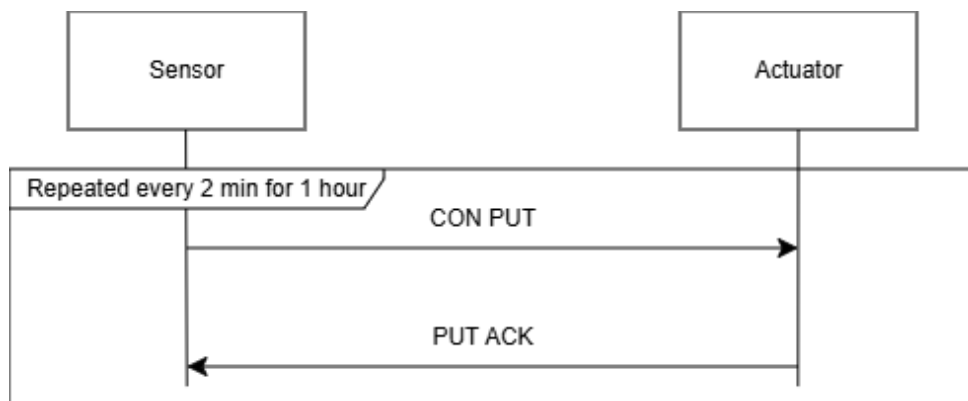
$$\text{Pub Ack} = 50 \text{ B} = 400 \text{ bit}$$

$$\text{Connect/Disconnect} = 60 \text{ B} = 480 \text{ bit}$$

$$\text{Connect Ack} = 50 \text{ B} = 400 \text{ bit}$$

### 1.1.1 EQ1a

For CoAP we have these message exchanges:



This is the consumption we calculated

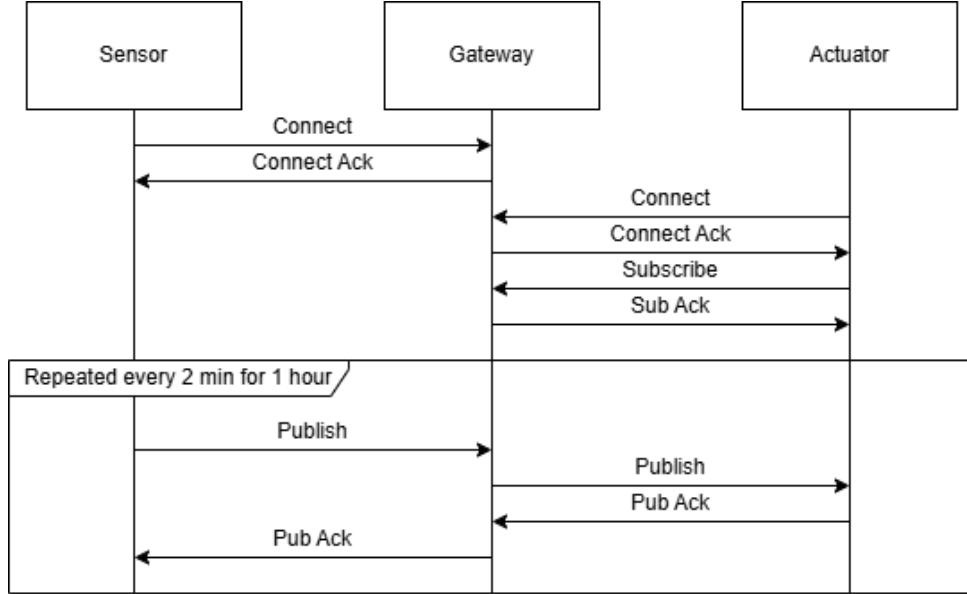
$$\begin{aligned}
 E_{\text{sensor}} &= E_{\text{tx}} \cdot (\text{CON PUT Request} \cdot 30) + E_{\text{rx}}(\text{PUT ACK} \cdot 30) = \\
 &= 50\text{nJ/bit} \cdot (560\text{bit} \cdot 30) + 58\text{nJ/bit} \cdot (120\text{bit} \cdot 30) = \\
 &= 840\mu\text{J} + 208.8\mu\text{J} = 1048.8\mu\text{J}
 \end{aligned}$$

$$\begin{aligned}
E_{\text{actuator}} &= E_{\text{tx}} \cdot (PUT\ ACK \cdot 30) + E_{\text{rx}}(CON\ PUT\ Request \cdot 30) = \\
&= 50\text{nJ/bit} \cdot (120\text{bit} \cdot 30) + 58\text{nJ/bit} \cdot (560\text{bit} \cdot 30) = \\
&= 180\mu\text{J} + 974.4\mu\text{J} = 1154.4\mu\text{J}
\end{aligned}$$

$$E_{\text{CoAP}} = E_{\text{sensor}} + E_{\text{actuator}} = 1048.8\mu\text{J} + 1154.4\mu\text{J} = 2203.2\mu\text{J}$$

### 1.1.2 EQ1b

For MQTT we have these message exchanges:



This is the consumption we calculated

$$\begin{aligned}
E_{\text{sensor}} &= E_{\text{tx}} \cdot (Connect + Publish \cdot 30) + E_{\text{rx}}(Connect\ Ack + Pub\ Ack \cdot 30) = \\
&= 50\text{nJ/bit} \cdot (480\text{bit} + 600\text{bit} \cdot 30) + 58\text{nJ/bit} \cdot (400\text{bit} + 400\text{bit} \cdot 30) = \\
&= 924\mu\text{J} + 719.2\mu\text{J} = 1643.2\mu\text{J}
\end{aligned}$$

$$\begin{aligned}
E_{\text{actuator}} &= E_{\text{tx}} \cdot (Connect + Subscribe + Pub\ Ack \cdot 30) + E_{\text{rx}}(Connect\ Ack + Sub\ Ack + Publish \cdot 30) = \\
&= 50\text{nJ/bit} \cdot (480\text{bit} + 520\text{bit} + 400\text{bit} \cdot 30) + 58\text{nJ/bit} \cdot (400\text{bit} + 440\text{bit} + 600\text{bit} \cdot 30) = \\
&= 650\mu\text{J} + 1092.72\mu\text{J} = 1742.72\mu\text{J}
\end{aligned}$$

$$E_{\text{MQTT}} = E_{\text{sensor}} + E_{\text{actuator}} = 1643.2\mu\text{J} + 1742.72\mu\text{J} = 3385.92\mu\text{J}$$

## 1.2 EQ2

I would choose MQTT with the Retain flag set to 1, so the gateway stores the latest value from the sensor and the actuator reads it upon waking up. This avoids the retransmissions we would have with direct CoAP, minimizing energy consumption.